Acute Bacterial Meningitis

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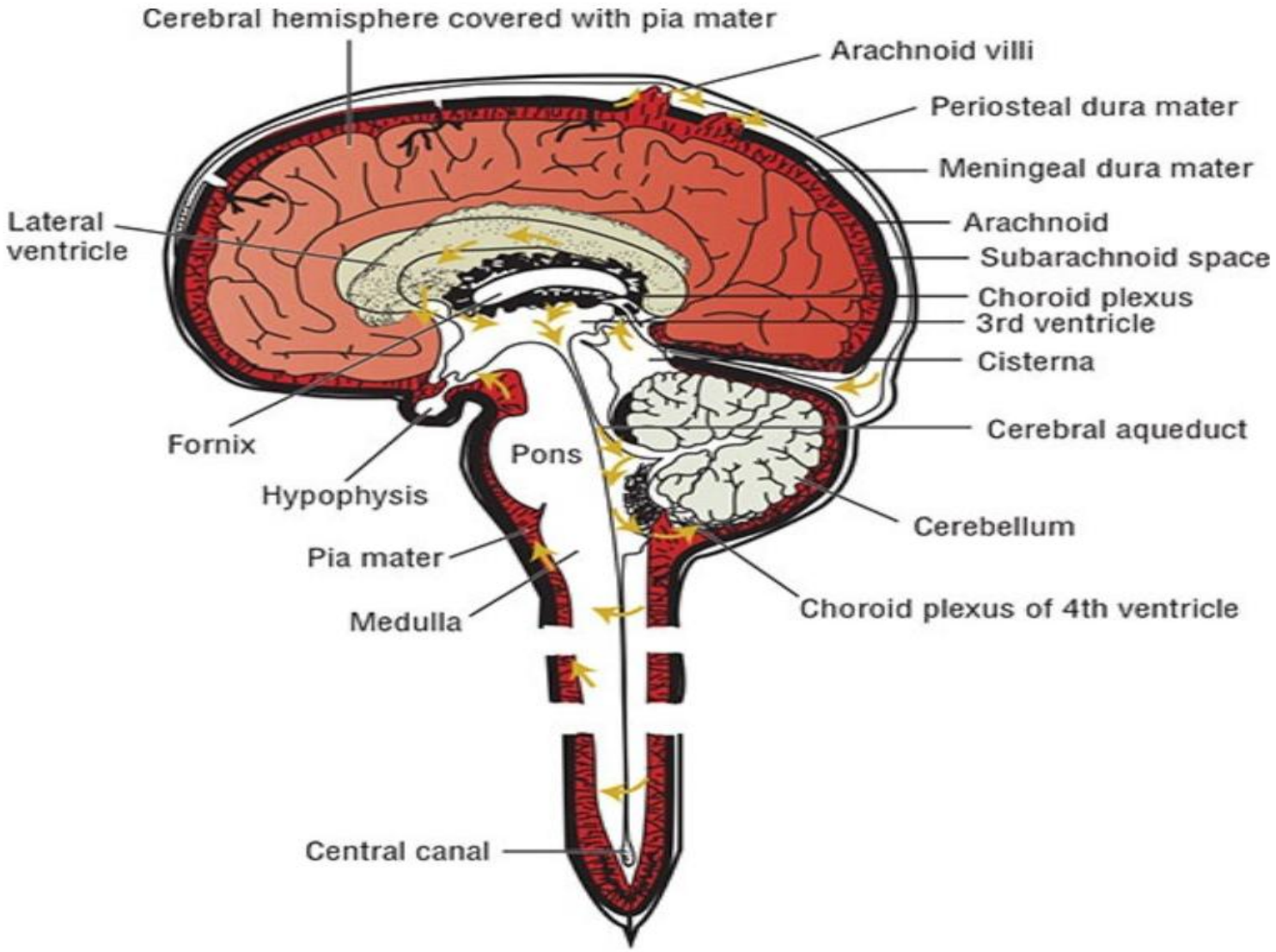
Definitions

- **Bacterial meningitis** is an acute purulent infection within the subarachnoid space
- **Meningoencephalitis**: involvement of meninges, SAS, and brain parenchyma
- **Aseptic meningitis**: Meningitis + no bacterial pathogen in CSF on Gram stain and routine culture

Anatomy and Physiology (1)

- **MENINGES**
  - Protective covering of the brain and spinal cord
  - Consist of three layers of fibrous tissue: the *pia mater, arachnoid,* and *dura mater*
  - Subarachnoid space contains CSF
  - Pia mater and arachnoid are known collectively as the *leptomeninges*
  - Dura mater is a tough outer membrane that adheres to the periosteum and vertebral column

Anatomy and Physiology(2)



Anatomy and Physiology (3)

- **CEREBROSPINAL FLUID**
  - Secreted by the choroid plexus in the lateral ventricles
  - Flows unidirectionally
  - About 550 mL/day of CSF is produced
  - Volume of CSF:
    - **Adults**: 150 mL,
    - **Children**: 60 to 100 mL, and
    - **Infants**: 40 to 60 mL

    ↳ To calculate **CSF** concentration of a drug administered intrathecally

Contents of Normal CSF

Type	Normal
WBC (cells/mm^3)	<5
Differential	Monocytes
Protein (mg/dL)	<50
Glucose (mg/dL)	45 – 80
Glucose (CSF/Serum) ratio	0.5 – 0.6
PH	7.3

Epidemiology

- Disease of the very young and very old
- Approximately 1.2 million cases every year worldwide
  - Resulting in ~170,000 deaths
- 20% of survivors will experience neurologic disabilities.
  - Seizures, sensorineural hearing loss, and hydrocephalus
- Risk Factors
  - Cigarette smoking
  - Cochlear implants
  - Others

Microbiology

- The bacterial causes of meningitis correlate well with;
  - Age and
  - Underlying conditions,
    - Head trauma
    - Recent neurosurgery
    - Alcoholism
    - Altered immune status

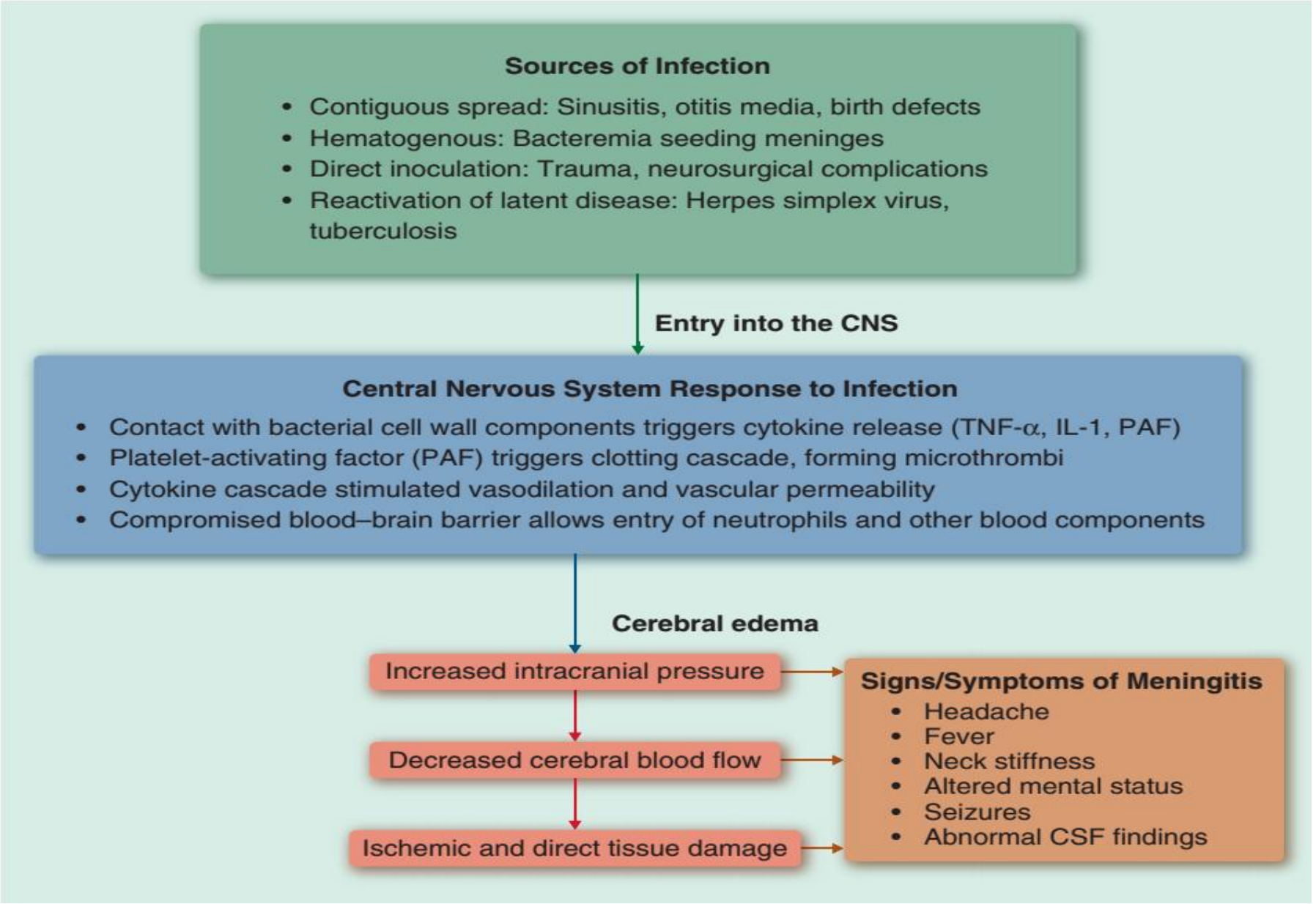
Microbiology

Age Group/Predisposing Condition	Most Likely Organisms
Neonates (<1 mo)	**Group B streptococcus** (S. agalactiae), E. coli, and other gram-negative bacilli (Klebsiella, Serratia species), Listeria monocytogenes
Infants and children (2–23 mons)	S. pneumoniae, N. meningitidis, S. agalactiae, H. influenzae, E. coli
Children and adults (2–50 yrs)	N. meningitidis, S. pneumoniae
Adults (>50 yr)	S. pneumoniae, N. meningitidis, L. monocytogenes, E. coli, Klebsiella species, and other g-vetive bacilli,
Postneurosurgical	S. aureus, gram-negative bacilli (E. coli, Klebsiella species, P. aeuruginosa), S. epidermidis
Closed head trauma	S. pneumoniae, H. influenzae, group A β-hemolytic streptococci
Penetrating trauma	S. aureus, S. epidermidis, gram-negative bacilli (E. coli, Klebsiella species, P. aeuruginosa)
Presence of risk factor (alcoholism and altered immune status)	S. pneumoniae, L. monocytogenes, H. influenzae, N. meningitidis

Pathophysiology

- **Meningitis** can develop from;
  - **Hematogenous spread** of organisms (the most common mechanism),
  - **Contiguous spread** from a parameningeal focus (e.g., sinusitis or otitis media), or
  - **Direct bacterial inoculation**, as occurs with head trauma or neurosurgery

Pathophysiology



Clinical Features

Signs and Symptoms of Acute Bacterial Meningitis	
✓ **Fever**	✓ Anorexia
✓ **Nuchal rigidity (stiff neck)**	✓ Headache
✓ **Altered mental status**	✓ Photophobia
✓ Seizures	✓ Nausea and vomiting
✓ Brudzinski's sign	✓ Focal neurologic deficits
✓ Kernig's sign	✓ Septic shock
✓ Irritability	

Clinical Features



Kernig's Signs



Brudzinski's signs

Lumbar puncture: CSF Findings

Type	Normal	Bacterial	Viral	Fungal	TB
WBC (cells/mm^3)	<5	1,000–5,000	100–1,000	40–400	100–500
Differential	Monocytes	Neutrophils	Lympho	Lympho	Variable
Protein (mg/dL)	<50	100–500	30–150	40–150	≤40–150
Glucose (mg/dL)	45 – 80	Low	Normal	Low	Low
Glucose (CSF/Serum) ratio	0.5–0.6	Decreased	Normal	Decreased	Decreased

Microbiologic Evaluation

- **Examination of**
  - CSF Gram stain
    - ✓Positive in more than 50% of cases
  - CSF culture
    - ✓It is always positive in purulent meningitis
    - ✓Can be negative with antecedent receipt of antibiotic therapy
  - Cultures from other sites (e.g., blood, sputum, urine)

TREATMENT

- **Desired Outcome**
  - Eradication of infection
  - Amelioration of signs and symptoms
  - Prevention of neurologic sequelae, such as seizures, deafness, coma, and death.

Treatment Principles

- Prompt initiation of appropriate antimicrobial therapy
  - ❖Delay in antibiotic administration is associated with increased morbidity and mortality
- The antibiotics selected must penetrate adequately into the CSF.
- The regimen chosen must have potent activity against known or suspected pathogens and
- Exert a **bactericidal** effect.

CSF Penetration Characteristics of Various Antimicrobials

Very Good

- ✓ Chloramphenicol, metronidazole, TMP-SMX, linezolid

Good

- ✓ Penicillins: Penicillin G, ampicillin, nafcillin, piperacillin, ticarcillin
- ✓ Other β-lactams: Aztreonam, clavulanic acid, imipenem, meropenem, sulbactam
- ✓ Cephalosporins: Cefepime, cefotaxime, ceftazidime, ceftizoxime, ceftriaxone, **cefuroxime**
- ✓ Fluoroquinolones: Ciprofloxacin
- ✓ Vancomycin, daptomycin
- ✓ Other agents: Rifampin

Fair to Poor

- ✓ Aminoglycosides: Amikacin, gentamicin, tobramycin
- ✓ Cephalosporines: 1st and 2nd generation
- ✓ Other agents: Azithromycin, clarithromycin, clindamycin, erythromycin

Empiric Therapy for Bacterial Meningitis

Age Group	Recommended Therapy	Alternative Therapy
Neonates (<1 mo)	Ampicillin + cefotaxime	Ampicillin + gentamicin
Infants and children (1month–2yrs)	Cefotaxime or ceftriaxone + vancomycin	Vancomycin + rifampin + aztreonam
Older children and adults (2–50 yr)	Cefotaxime or ceftriaxone + vancomycin	Vancomycin + rifampin + aztreonam
Elderly (>50 yr)	Ampicillin + cefotaxime, or ceftriaxone + vancomycin	Vancomycin + TMP-SMX + aztreonam

Empiric Therapy

Predisposing Condition	Recommended Therapy	Alternative Therapy
Post-neurosurgical	Vancomycin + ceftazadime	Vancomycin + cefepime or meropenem
Closed head trauma	Cefotaxime or ceftriaxone + vancomycin	Vancomycin + rifampin + aztreonam
Penetrating head trauma	Vancomycin + ceftazadime	Vancomycin + cefepime or meropenem
Presence of risk factor (alcoholism and altered immune status)	Vancomycin + ceftriaxone or cefotaxime + ampicillin	Vancomycin + TMP-SMX + aztreonam

Antibiotic Dosing Regimens

Antibiotic	Daily Dose (interval in hours)[a]			
	Neonates			
	0–7 days Old	8–28 days Old	Infants and Children	Adults
Ampicillin	150 mg/kg (8)	200 mg/kg (6–8)	300 mg/kg (6)	12 g (4)
Aztreonam				8 g (6)
Nafcillin	75 mg/kg (8–12)	100–150 mg/kg (6–8)	200 mg/kg (6)	12 g (4)
Penicillin G	0.15 million units/kg (8–12)	0.2 million units/kg (6–8)	0.3 million units/kg (4–6)	24 million units (4)
Meropenem			120 mg/kg (8)	6 g (8)
Cephalosporins				
Cefotaxime	100–150 mg/kg (8–12)	150- 200 mg/kg (6–8)	225–300 mg/kg (6–8)	12 g (4)
Ceftriaxone			80–100 mg/kg (12–24)	4 g (12)
Cefazidime	100–150 mg/kg (8–12)	150 mg/kg (8)	150 mg/kg (8)	6 g (8)
Cefepime			150 mg/kg (8)	6 g (8)
Aminoglycoside[b,c]				
Gentamicin	5 mg/kg (12)	7.5 mg/kg (8)	7.5 mg/kg (8)	5–7 mg/kg (24)
Tobramycin	5 mg/kg (12)	7.5 mg/kg (8)	7.5 mg/kg (8)	5–7 mg/kg (24)
Amikacin	15–20 mg/kg (12)	30 mg/kg (8)	20–30 mg/kg (8)	15 mg/kg (24)
Linezolid				1,200 mg (12)
Rifampin		10–20 mg/kg (12)	10–20 mg/kg (12–24)	600 mg (24)
TMP-SMX[d]			10–20 mg/kg (6–12)	10–20 mg/kg (6–12)
Vancomycin[c,e]	20–30 mg/kg (8–12)	30–45 mg/kg (6–8)	60 mg/kg (6)	30–45 mg/kg (8–12)

Adjunctive Corticosteroid Therapy

- Corticosteroids can reduce cerebral edema and lower intracranial pressure.
  - Reduce the synthesis and release of the proinflammatory cytokines; TNF-α and IL-1β from monocytes and astrocytes.
    - Initiate the cascade of events that lead to neuronal tissue damage and neurologic sequelae.
  - Significantly reduced hearing loss and neurologic sequelae
  - Dexamethasone 0.15 mg/kg every 6 hours for 2 to 4 days,
    - 20 minutes prior to or concomitant with the first dose of antimicrobials

Definitive Therapy

- ***Haemophilus influenza***
  - β-Lactamase-negative: Ampicillin
    - ✓Alternative: Cefotaxime or ceftriaxone; aztreonam
  - β-Lactamase-positive: Cefotaxime or ceftriaxone
    - ✓Alternative: Aztreonam
- ***Neisseria meningitidis***
  - Cefotaxime or ceftriaxone
  - Alternative: Meropenem
- ***Streptococcus pneumoniae***
  - Cefotaxime or ceftriaxone
  - Alternative: Vancomycin

Definitive Therapy

- ***Pseudomonas aeruginosa***
  - ✓Cefepime or ceftazidime; meropenem
  - ✓**Alternative**: Aztreonam
- ***Staphylococcus aureus***
  - Methicillin-susceptible (MSSA): Nafcillin or oxacillin
    - ✓**Alternative**: Vancomycin ± rifampin; meropenem; linezolid
  - Methicillin-resistant (MRSA): Vancomycin ± rifampin
    - ✓**Alternative**: TMP-SMX; linezolid
- ***Staphylococcus epidermidis*:**
  - ✓Vancomycin ± rifampin
  - ✓**Alternative**: TMP-SMX; linezolid

Definitive Therapy

Streptococcus agalactiae

- ✓Penicillin G or ampicillin + gentamicin
- ✓**Alternative**: Cefotaxime or ceftriaxone

Listeria monocytogenes

- ✓Penicillin G or ampicillin ± gentamicin
- ✓**Alternative**: TMP-SMX

Enterobacteriaceae

- *E. coli, Klebsiella*: Cefotaxime or ceftriaxone
  - ✓**Alternative**: Cefepime; aztreonam; meropenem
- *Enterobacter, Serratia*: Cefepime; meropenem
  - ✓**Alternative**: TMP-SMX; aztreonam

Duration of Therapy

Etiology	Duration
Haemophilus influenzae	7
Neisseria meningitidis	7
Streptococcus pneumoniae	10 – 14
GBS (S. agalactiae)	14 – 21
Listeria monocytogenes	>21
Gram-negative bacilli (P. auroginosa)	21

Follow-up and Monitoring

- **Clinical signs and symptoms (every 4 hours)**
  - Fever, altered mental status, and stiff neck,
- **Laboratory tests**
  - CBC with differential
  - Serum electrolytes (Na, K, Cl, HCO3),
  - Blood glucose, and
  - Renal function tests (e.g., BUN, SCr)
- **Microbiologic Findings**
  - CSF and blood samples for Gram stain, cultures, and sensitivity testing
- **CSF Examination**

Quiz

1. Which of the following aetiology of meningitis need steroid therapy

A) H. influenza B) L. monocytogen C) S. auras D) All

2. Timing of dexamethasone administration in pyogenic meningitis

A) At any time of antibiotic administration

B) At the same time or with in 20 minute of Abx administration

C) 1hr before antibiotic administration

D) All

3. 70 years old RVI patient is diagnosed with bacterial meningitis. He has history of penetrating head trauma 3 weeks back. What is the best empiric therapy for this patient?

A) Ampicillin + vancomycin + ceftazidime

B) Ampicillin +ceftriaxone +vancomycin

C) Ceftriaxone + vancomycin

D) Ampicillin +ceftriaxone

Quiz …….cont'd

4. What is the duration of antibiotic therapy for patients diagnosed with P. aeruginosa associated meningitis?

A) 7 days B) 14 days C) 10 days D) 21 days

5. True about CSF analysis of bacterial meningitis

A) Increase both protein and glucose

B) Increase neutrophils count and glucose

C) Increase protein and decrease glucose

D) Increase monocyte count and decrease protein

Thanks